



Secant Varieties of Flag Varieties

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1. Secants & Dimension

The study of **Secant Varieties** bridges projective geometry and tensor decomposition.

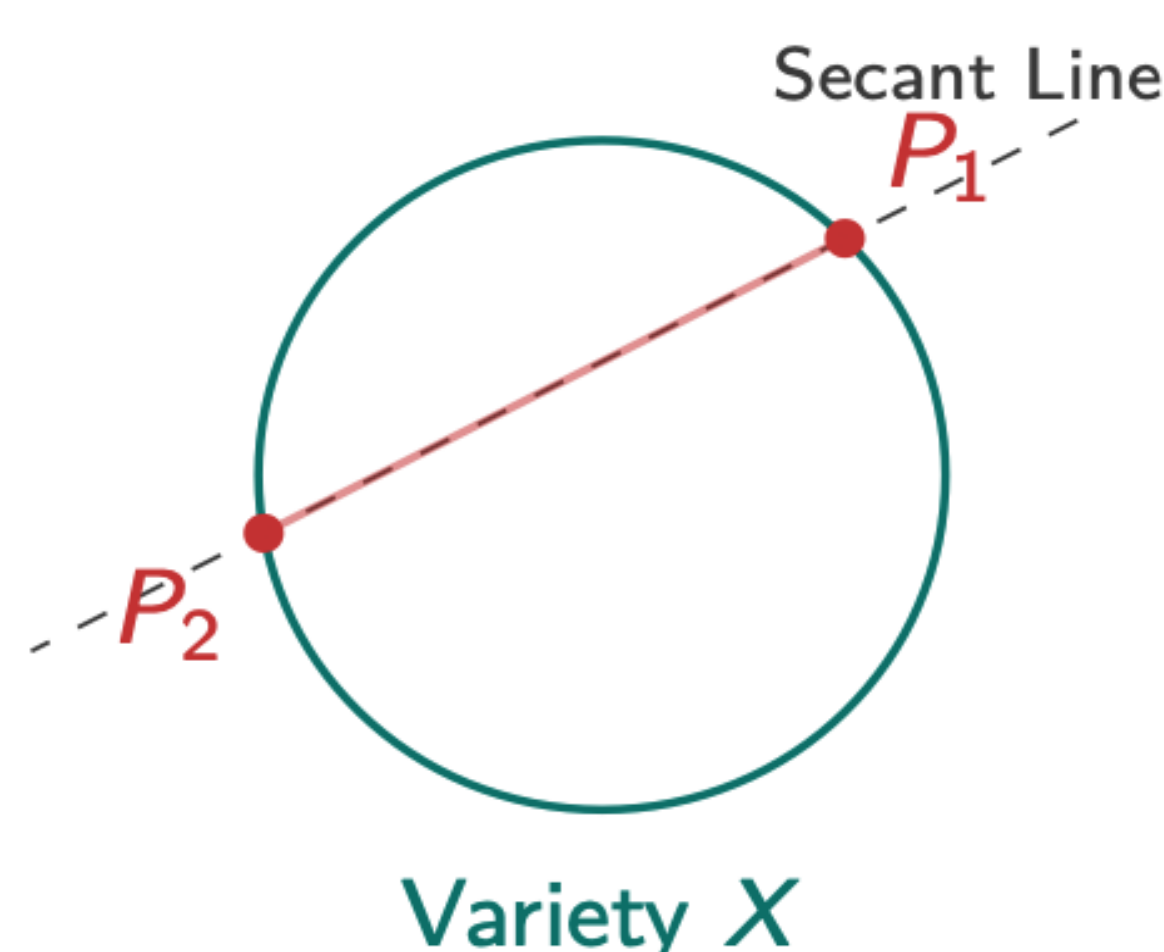
- Let $X \subset \mathbb{P}^N$ be a projective variety. The s -th secant variety is the closure of the union of spans of s points:

$$\sigma_s(X) = \overline{\bigcup_{P_1, \dots, P_s \in X} \langle P_1, \dots, P_s \rangle}.$$

- Our problem:** compute $\dim \sigma_s(X)$.
- Terracini's Lemma** relates the tangent space of the secant variety to the span of tangent spaces at general points:

$$\widehat{T}_{\sigma_s(X)} = \langle \widehat{T}_{P_1} X, \dots, \widehat{T}_{P_s} X \rangle.$$

- Defectivity:** We define $\exp.\dim(\sigma_s(X))$ as $\min\{N, s(\dim X + 1) - 1\}$. If $\dim(\sigma_s(X)) < \exp.\dim(\sigma_s(X))$, then X is said to be *defective*.



2. The Dual Terracini Lemma

- For **Veronese varieties** (rank-1 symmetric tensors), we have **Lasker's Lemma**:

$$(\widehat{T}_P X)^\perp = (I^2(p))_d.$$

- We get to the **Dual Terracini Lemma** → **interpolation problem** (A-H theorem).

$$(\widehat{T}_{\sigma_s(X)})^\perp = I^2(p_1)_d \cap \dots \cap I^2(p_s)_d.$$

- The Goal:** Extend this “Dual Terracini” framework to **flag varieties**.

Let $\lambda = (\lambda_1^{a_1}, \dots, \lambda_k^{a_k})$, $\lambda' = (n_k^{d_k}, \dots, n_1^{d_1})$. We have $\mathbb{F}(n_1, \dots, n_k; V)$ embedded in the Schur module $\mathbb{P}(\mathbb{S}_\lambda V)$ via $\mathcal{O}(d_1, \dots, d_k)$.

$$\mathbb{F} \hookrightarrow \mathbb{P}(\mathbb{S}_\lambda V) \implies (\widehat{T}_P \mathbb{F})^\perp \stackrel{?}{=} I_{\lambda'}^2(P)$$

3. The Naive Approach

- Natural guess:** Is the apolar tangent space the square of the **Schur Apolar Ideal**?

$$I_S(P) = \{f \in \mathbb{S}^\bullet V^* \mid f * P = 0\} \subset (\mathbb{S}^\bullet V, \mathcal{M}_V^{\lambda, \mu}).$$

We proved it for Grassmannians, but we showed that it fails for general flags.

- Counterexample:** $\mathbb{F}(1, 2; \mathbb{C}^3)$ embedded in $\mathbb{P}(\mathbb{S}_{(4,2)} V)$ via $\mathcal{O}(2, 2)$. Let P be the standard flag.

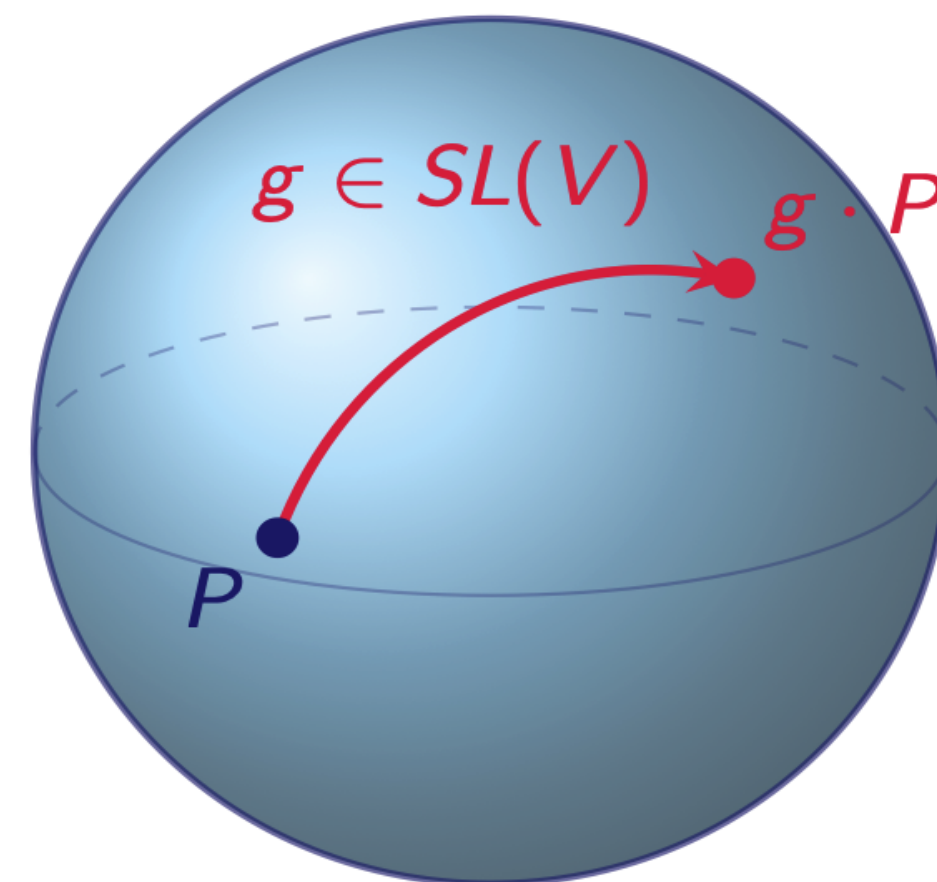
$$\tilde{x} = (x_1 \wedge x_2)^{\otimes 2} \otimes x_2^{\otimes 2}$$

- Geometrically:** $\tilde{x} \in (\widehat{T}_P F)^\perp$ (it vanishes on the tangent space).
- Algebraically:** $\tilde{x} \notin I_S^2(P)_\lambda$.

To generate $\tilde{x}$ in the square ideal, we would need factors in $I_S(P)$ that do not exist.

4. The Solution: Jet Ideal

We define an ideal using the intrinsic geometry of $F \simeq SL(V)/\text{Stab}(P)$.



We will use the Lie algebra $\mathfrak{g} = \mathfrak{gl}(V)$.

- The group $GL(V)$ acts on $\mathbb{S}^\bullet V^*$.
- Differentiating this action gives an action of $\mathfrak{gl}(V)$ as **derivations**. Let $\xi \in \mathfrak{gl}(V)$ be a derivation.

Definition (Jet Ideal of order 2).

$$I^{(2)}(P) := \{f \in \mathbb{S}^\bullet V^* \mid f * P = 0 \text{ and } (\xi \cdot f) * P = 0\}$$

where $\xi \cdot f$ is the derivative of f along the flow generated by ξ .

This is the correct object. We proved:

Theorem 1 (Dual Terracini for Flags, Isoldi 2025). For any flag variety embedded in a Schur module as the minimal orbit,

$$(\widehat{T}_P F)^\perp = I^{(2)}(P)_\lambda.$$

This restores a correspondence similar to the Veronese case.

5. A Computational Tool

Computing derivations and multiplying in Schur modules is **expensive**. We map the problem to a multi-graded polynomial algebra. Let λ' be the conjugate partition with parts n_i .

Definition (Slot-by-Slot Algebra).

$$\mathcal{R}_\lambda := \bigotimes_{i=1}^k \text{Sym}^\bullet (\wedge^{n_i} V^*)$$

This algebra is graded by multidegrees $\mathbf{a}$. We utilize the $GL(V)$ -equivariant projection:

$$\pi_\lambda : (\mathcal{R}_\lambda)_\mathbf{d} \longrightarrow \mathbb{S}_\lambda V^*.$$

On $\mathcal{R}_\lambda$, we define the ideal corresponding to the point P (representing the flag $W_1 \subset \dots \subset W_k$).

Definition (Slot Ideal). The slot ideal $I_{\text{slot}}(P) \subset \mathcal{R}_\lambda$ is the ideal generated in multidegree $(0, \dots, 1, \dots, 0)$ by the annihilators in each tensor slot:

$$\left(\{1 \otimes \dots \otimes (\wedge^{n_i} W_i)^\perp \otimes \dots \otimes 1 \mid i = 1, \dots, k\} \right).$$

Example: Let $P = [(e_1 \wedge e_2)^{\otimes 2} \otimes e_1^{\otimes 2}] \in \mathbb{F}(1, 2; V) \hookrightarrow \mathbb{P}(\mathbb{S}_{(4,2)} V)$.

$$I_{\text{slot}}(P) = ((x_2 \otimes 1), (x_3 \otimes 1), (1 \otimes x_1 \wedge x_3), (1 \otimes x_2 \wedge x_3)).$$

6. The Consistency Theorem

One of our **main results**: we proved that under some conditions, the computationally simple Slot Ideal recovers the complex Jet Ideal.

Theorem 2 (Consistency, Isoldi 2025).

Let $\mathbf{d}$ be the embedding degrees. If $d_i \geq 2$ for all i , then:

$$\pi_\lambda ((I_{\text{slot}}(P)^2)_\mathbf{d}) = I^{(2)}(P)_\lambda$$

Implication: We can compute dimensions of secant varieties using standard Gröbner basis algorithms on $I_{\text{slot}}(P)$ in the polynomial ring $\mathcal{R}_\lambda$.

7. Experimental Results

We implemented this framework in Macaulay2 and tested the defectivity/regularity of several flag varieties.

Variety	s	Exp.	Dimension
$\mathbb{G}(2, 3), \mathcal{O}(2)$	2	5	4 (Defect.)
$\mathbb{G}(2, 3), \mathcal{O}(2)$	3	5	5 (Fills)
$\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$	6	23	23 (Reg.)
$\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$	7	26	25 (Defect.)
$\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$	8	26	26 (Fills)
$\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(1, 1)$	2	7	6 (Defect)
$\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(1, 1)$	3	7	7 (Fills)
$\mathbb{F}(1, 2, 3; \mathbb{C}^4), \mathcal{O}(1, 1, 1)$	2	13	13 (Reg.)

8. Conclusions & Future Work

Key Contributions:

- Dual Terracini for Grassmannians** in the Schur-theoretic setting.
- Dual Terracini for flag varieties** via the Jet Ideal and Lie algebra derivations.
- Consistency Theorem** linking Jet Ideal and Slot-by-Slot algebra.
- Computational framework** for secant dimensions of flag varieties.

Open Questions:

- Can this method lead to a **classification** of defective secant varieties of flags?
- What happens when embedding degrees $d_i < 2$?
- Which partitions allow Plücker relations to resolve ambiguities?

Practical Impact: Complex Schur-module computations are replaced by polynomial algebra, opening the door to a systematic computational study of the geometry of the secants of flag varieties.

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